Edinburgh Napier UNIVERSITY

Visualising Tabular Data

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SET09123 – Interactive Data Visualisation

Lecture Summary

- Tabular Data
- Choosing a Visualisation
- Common Visualisations

Tabular Data

How would you define Tabular Data?

Tabular Data

- Tabular Data is data that can be expressed in the form of a *table*
  - Data written across *rows* and *columns*

Tabular Data

- Each *row* represents an item in the data, or entry

Item 0	…	…	…	…	…
Item 1	…	…	…	…	…
Item 2	…	…	…	…	…
…	…	…	…	…	…

Tabular Data

- Each *column* represents an attribute of the data, or dimension

Attr 0	Attr 1	Attr 2	Attr 3	Attr 4	Attr 5
Item 0	…	…	…	…	…
Item 1	…	…	…	…	…
Item 2	…	…	…	…	…
…	…	…	…	…	…

Tabular Data

- Each *cell* holds the value for one attribute of one item

Title	Author	Genre	Inventory	Unit Price	Attr 5
The Illiad	Homer	Epic	6	3.99	…
Death on…	Christie	Crime	5	5.99	…
The Hobbit	Tolkien	Fantasy	7	6.99	…
…	…	…	…	…	…

Tabular Data

- Some attributes *uniquely identify* items: key attributes

Title	Author	Genre	Inventory	Unit Price	ISBN
The Illiad	Homer	Epic	6	3.99	978014…
Death on…	Christie	Crime	5	5.99	978140…
The Hobbit	Tolkien	Fantasy	7	6.99	978004…
…	…	…	…	…	…

Visualising Tabular Data

- Visualising tabular data means:

  1. Creating a mark for each item
  2. Encoding chosen attributes with visual channels

Visualising Tabular Data

- But before that, we must:
  - Understand our data
    - the data types, the number of dimensions
  - Determine what to show
    - e.g., comparison, correlation, distribution, composition

Choosing a Visualisation

Choosing a Visualisation

- There are numerous guides out there to help you choose a visualisation
  - E.g., datavizcatalogue.com

- I strongly recommend datavizproject.com
  - I will use their examples in these slides (CC-BY-NC-ND)

Visualisation Functions

- We will go through 4 of the main visualisation functions:
  - Comparison
  - Correlation
  - Distribution
  - Composition

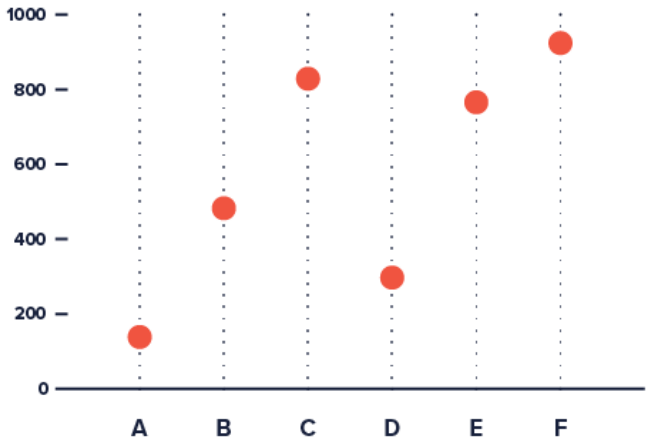
- Note that the examples shown can achieve several of these functions, not just the one I put them into
  - E.g., an histogram shows the *distribution* of one attribute <u>and</u> enables *comparison* between bins

Comparison

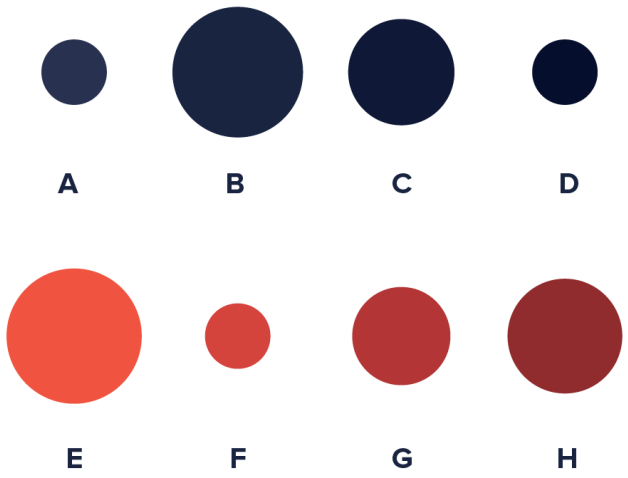
When you want to enable the comparison between items or values

- Comparison will involve a numerical value
  - Sometimes it is a derivative value, e.g., count of items per categories
- Comparison alone typical means between categories
  - Correlation, distribution and composition will imply comparison

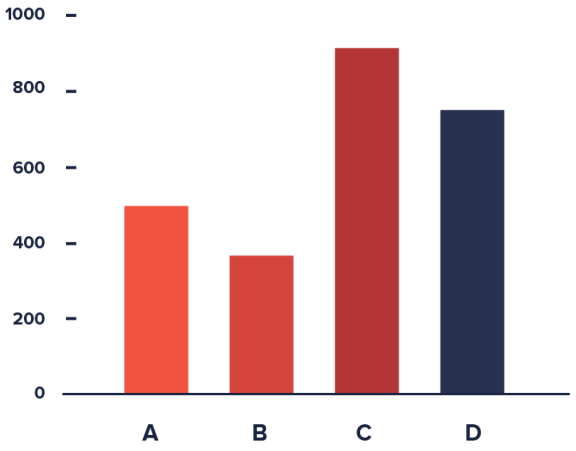
Comparison – Across Categories



Dot Chart



Proportional Area



Bar Chart

Image Source: datavizproject.com

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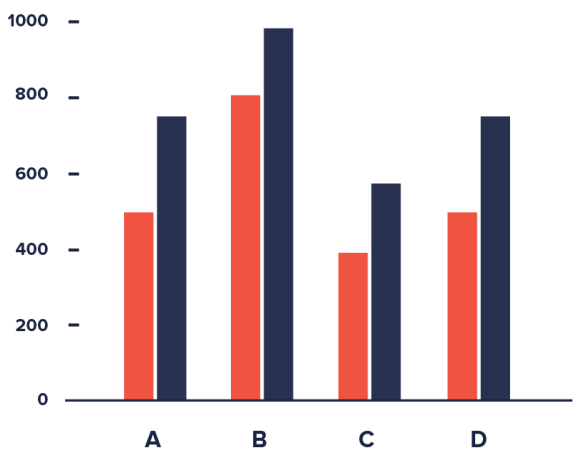
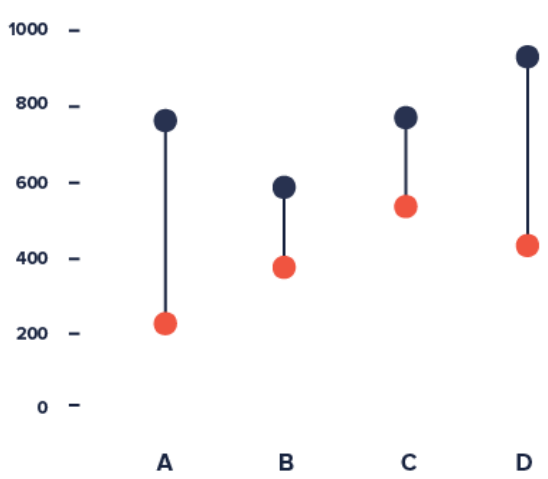
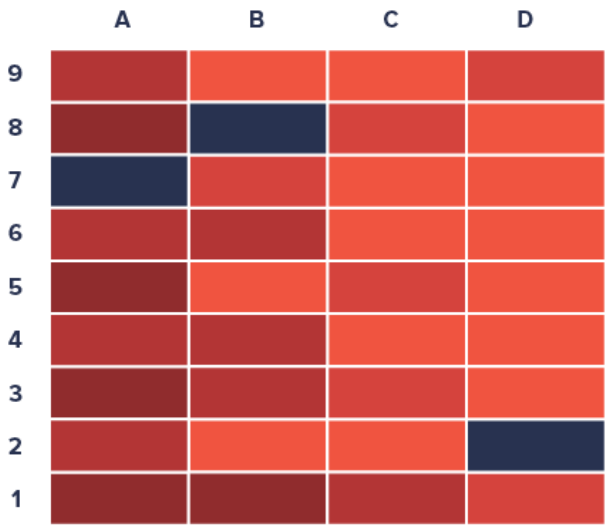
Comparison – More Categories



Grouped Bar Chart



Dumbbell Plot



Heat Map

Image Source: datavizproject.com

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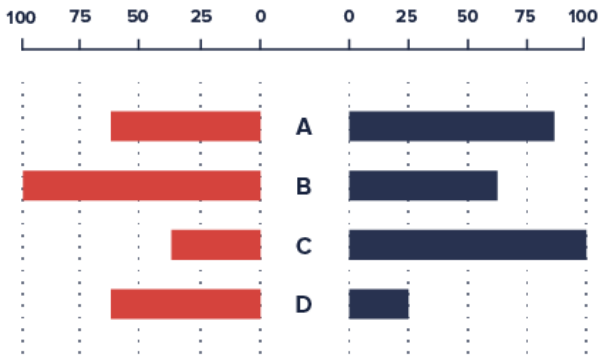
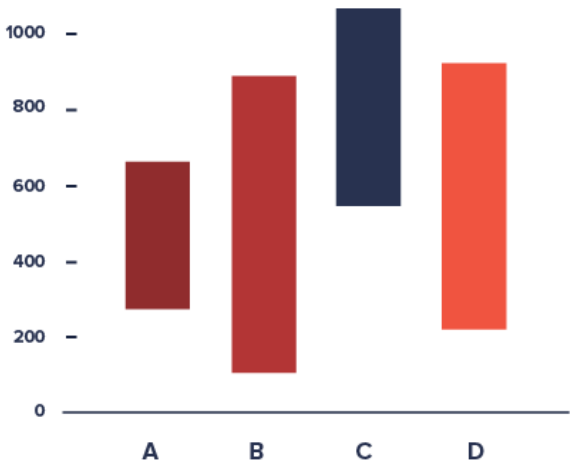
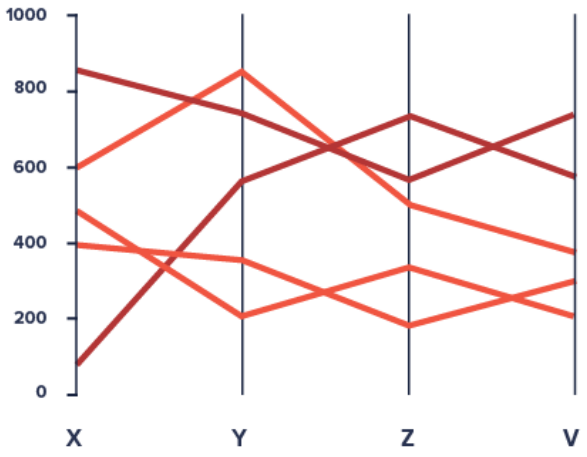
Comparison – More Values



Butterfly Chart



Span Chart



Parallel Coordinates

Image Source: datavizproject.com

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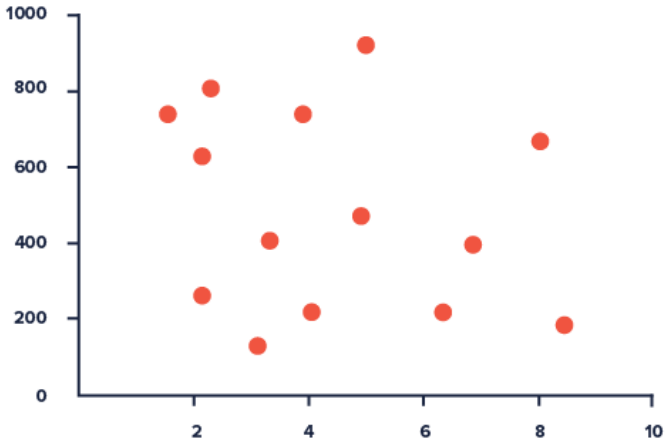
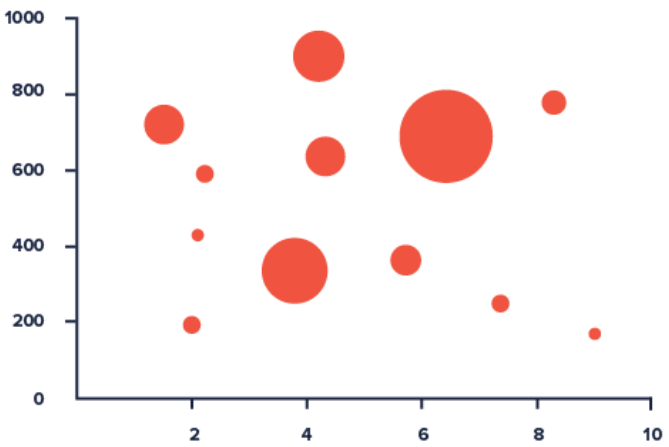
Correlation

When you want to show the relationship between items or attributes

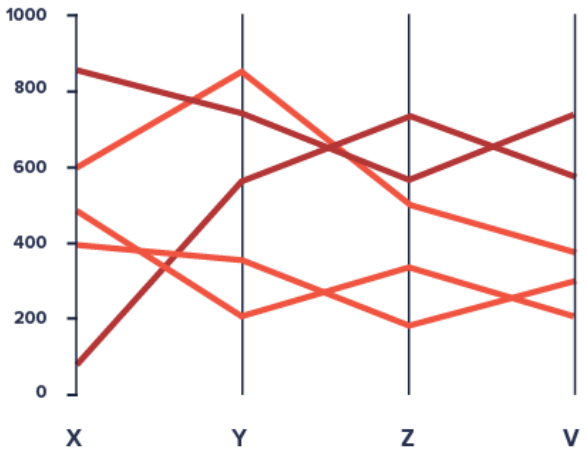
Correlation – Between Numbers



Scatter Plot



Bubble Chart



Parallel Coordinates

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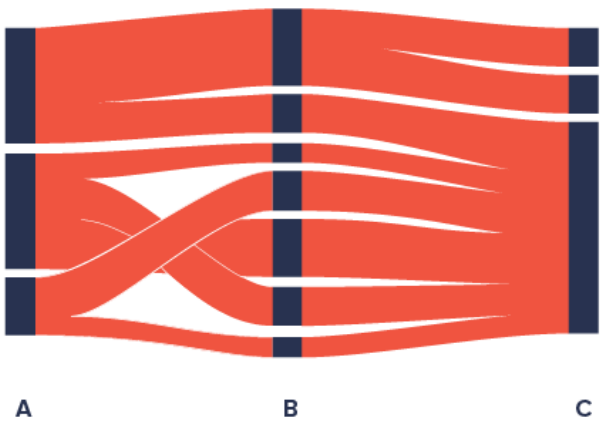
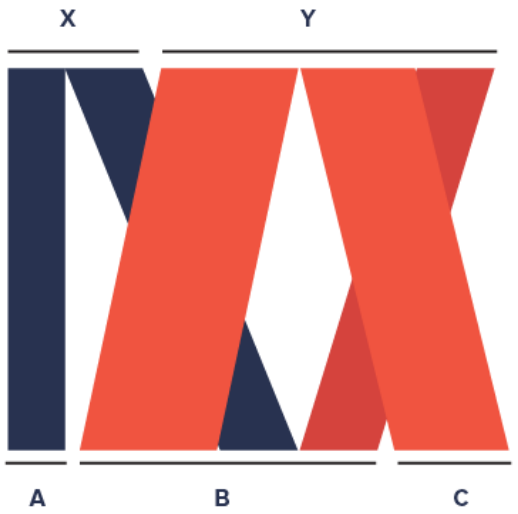
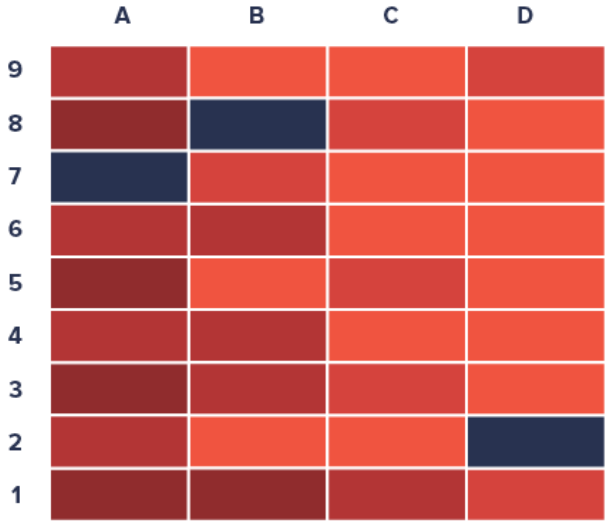
Correlation – Between Categories



Alluvial Diagram



Parallel Sets



Heat Map

Image Source: datavizproject.com

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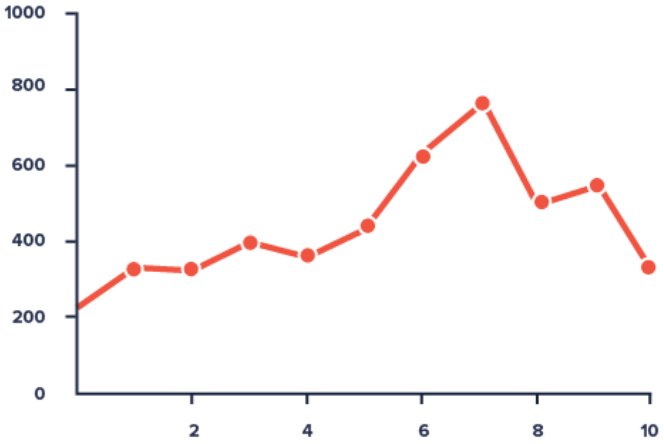
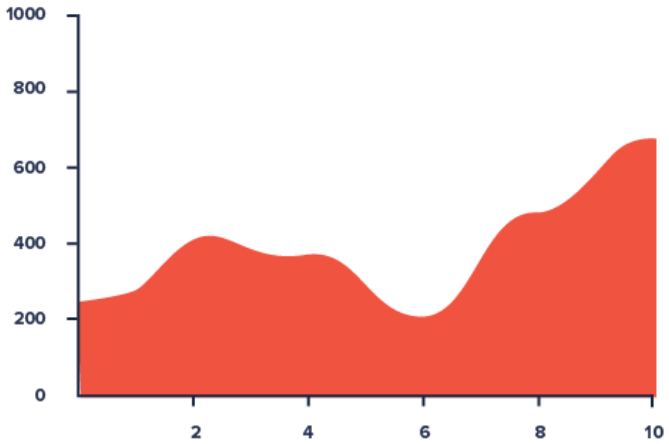
Distribution

When you want to display the spread of items or values

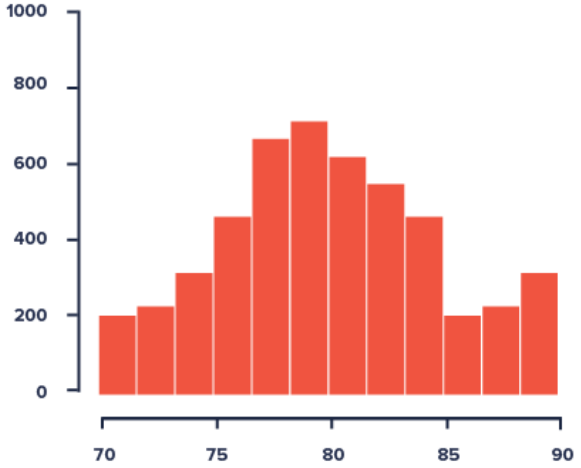
Distribution – One Category



Line Chart



Area Chart



Histogram

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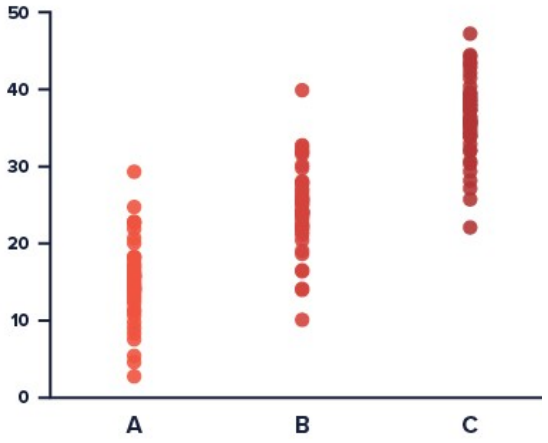
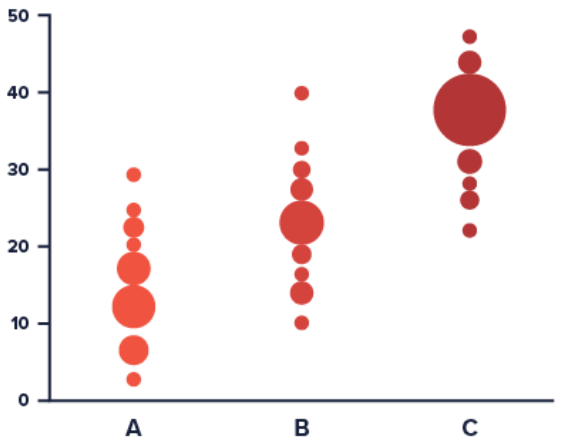
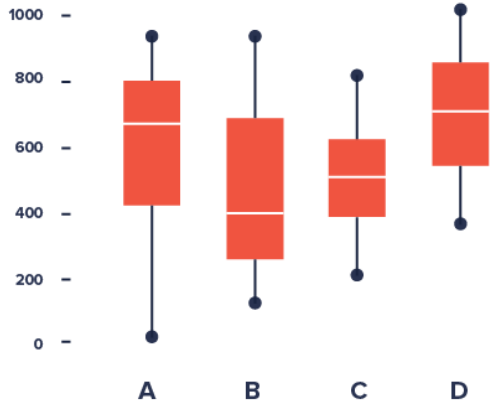
Distribution – Multiple Categories



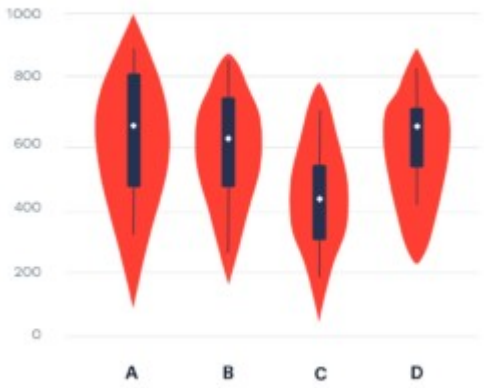
Strip Plot



Count Plot



Box Plot



Violin Plot

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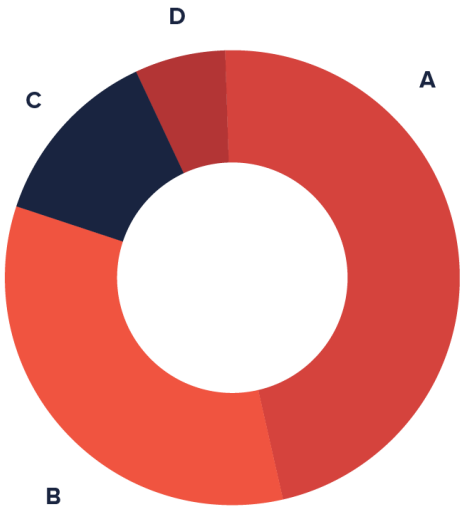
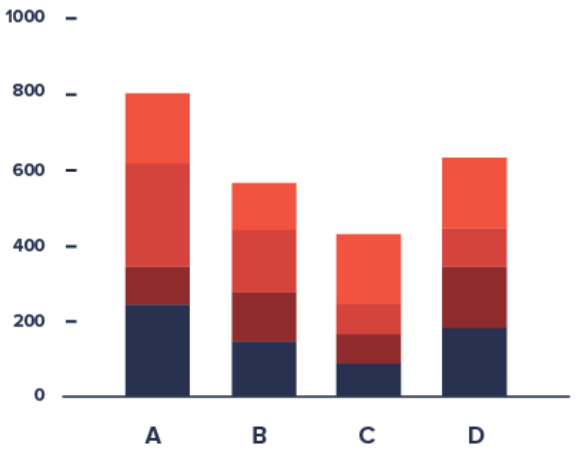
Composition

When you want to represent a value as part of a bigger whole

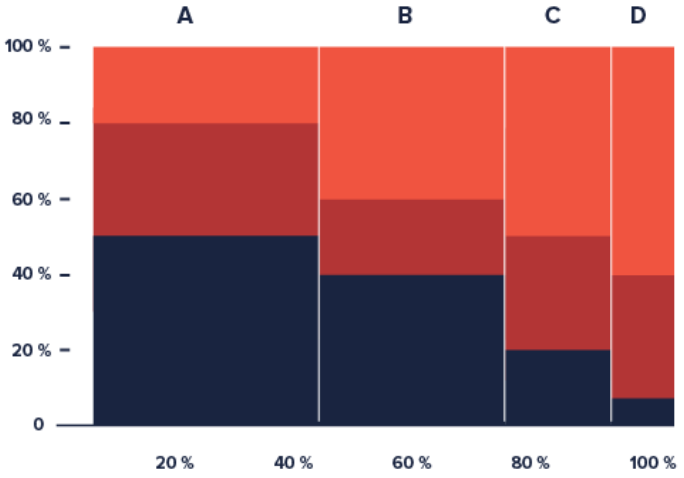
Composition



Pie/Donut Chart



Stacked Bar Chart



Marimekko Chart

Image Source: datavizproject.com

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Conclusion

Key Points

- Tabular data is data that can be formatted as a table:
  - Rows are *items* or entries
  - Columns are *attributes* or dimensions

- Choosing a visualisation relies on:
  - Understanding the data: types of attributes, number of dimensions
  - Determining the desired function

- There are 4 main functions: comparison, correlation, distribution and composition